

ASIMOV v1

PRELIMINARY SAFETY CASE — EXECUTIVE SUMMARY

Open-source humanoid · 1.2 m · 35 kg · 25 actuated DOF · Doc ID ASV1-SC-CP-001 r0.1 · 2026-05-03 · github.com/jherrodthomas/asimov-v1

Important — This is a preliminary, concept-phase safety case. It does NOT certify Asimov v1 as safe to deploy. The upstream release publishes mechanical CAD, electrical schematics and a MuJoCo model only — no HARA, FMEA, FMEDA, controller code, E-Stop validation, or BMS specification is published. Provisional acceptance applies only to a supervised, tethered laboratory setting (OE-1).

Hazards	Safety functions	Standards	Open gaps	Very-high gaps	Acceptance
32	25	5+3 ref	28	11	OE-1 only

Standards framework applied

Standard	Coverage
ISO 12100:2010	Master risk-assessment method
ISO 10218-1/-2:2011 + ISO/TS 15066:2016	Industrial robot envelope; collaborative-robot PFL
ISO 13849-1:2023 / -2:2012	Safety-related parts of control systems — PL determination & validation
ISO 13482:2014	Personal-care robot — Type 1 mobile servant hazard catalogue
ISO 9001:2015	Quality management — design control, NCR, CAR
IEC 60204-1, IEC 62133-2 (referenced)	Electrical equipment of machines; Li-ion battery safety

Risk profile by operational envelope

Envelope	Description	Risk	Provisional acceptance
OE-1 Lab (supervised)	Trained operator, tethered, no bystanders	LOW	PROVISIONAL YES
OE-2 Industrial cobot	Trained users, defined cell	MEDIUM	NOT YET — 12 gaps
OE-3 Home / consumer	Untrained users, mixed surfaces, children/pets	HIGH	NOT YET — 19 gaps
OE-4 Public space	Bystanders, weather, kerbs	VERY HIGH	NOT YET — 21 gaps

Top 5 residual hazards (after concept-level safety functions)

Hazard	Scenario	PLr	Residual	Reason
HZ-T02	Li-ion thermal runaway (battery)	d	VERY HIGH	BMS user-supplied; not gated
HZ-P02	Locomotion policy edge case	e	HIGH	Policy unreleased; sim2real gap
HZ-P04	Intentional contact exceeds force limit	e	HIGH	PFL thresholds not declared
HZ-M02	Joint pinch during MOVE	e	MED	Joint speed limits not declared
HZ-M04	Arm sweep strike	e	MED	No speed reduction trigger yet

Top 12 gaps to close (of 28 total)

ID	Standard / clause	Description	Sev
GAP-13482-04	ISO 13482 §5.16	Wireless E-Stop architecture, latency, link surveillance	VHIGH
GAP-10218-03	ISO 10218-1 §5.5.3	E-Stop architecture (Cat 0/1, dual-channel)	VHIGH
GAP-13849-02	ISO 13849-1 §7	MTTFD/DC/CCF figures absent — PL cannot be computed	VHIGH
GAP-13849-03	ISO 13849-2	Validation plan & test report missing	VHIGH
GAP-13482-01	ISO 13482 §5.7	Force/energy limits not declared	VHIGH
GAP-13482-05	ISO 13482 §6	Safety-related control-system implementation gaps	VHIGH
GAP-13482-06	ISO 13482 §7	V&V plan and report not produced	VHIGH
GAP-13849-01	ISO 13849-1 §6	Cat 3 redundancy not implemented for 10 SFs	VHIGH
GAP-BMS-01	IEC 62133-2	Battery user-supplied — no compliance gate	VHIGH
GAP-LP-01	(this case)	Locomotion policy unreleased	VHIGH
GAP-12100-01	ISO 12100 §6.4	Adequacy-of-risk-reduction record absent	HIGH
GAP-DOC-01	(this case)	Manual /v1/safety, /v1/quickstart, /v1/bom return 404	HIGH

Recommended next-revision actions

1	Publish wireless E-Stop architecture (Cat 1, dual-channel, link surveillance, fault-reaction time) per ISO 13849-1 / ISO 13482 §5.16.
2	Declare joint torque, velocity and momentum limits per joint with verification report — ISO 13482 §5.10.
3	Run a power-and-force-limiting (PFL) measurement campaign per ISO/TS 15066 if cobot use is intended.
4	Compute and publish MTTFD, DC and CCF for each safety-rated control channel — ISO 13849-1 §7.
5	Issue an ISO 13849-2 validation plan and test report covering the 25 safety functions.
6	Gate the DIY kit against pre-qualified Li-ion packs that meet UN 38.3 / IEC 62133-2 with a compliant BMS.
7	Restore manual subpages /v1/safety, /v1/quickstart, /v1/bom with explicit operating envelope and warnings — ISO 12100 §6.5.3.
8	Stand up an ISO 9001-aligned QMS subset: design-review records, NCR/CAR procedure, supplier control, internal audit.
9	Verify the locomotion policy against the 32-hazard catalogue; publish sim-to-real evidence including domain randomisation and worst-case-latency budgets.
10	Specify and authenticate the tele-op channel (TLS, token, integrity protection, idle-timeout) before the Asimov API ships.

Conclusion. Provisional acceptance for OE-1 (research lab, supervised, tethered) only. Deployment to industrial cobot, home/consumer, or public-space environments is BLOCKED pending closure of the gaps above. Closing the top 12 unlocks OE-1 unconditionally and OE-2 conditionally; closing all 28 plus regulatory engagement is required for OE-3/OE-4.

Companion artefacts: **Asimov_v1_Safety_Case.xlsx** (15-tab workbook with HARA, FMEA, FSC, GSN, evidence and gap registers) and **Asimov_v1_Safety_Case_Report.docx** (full narrative report). Source repository: github.com/jherrodthomas/asimov-v1.